

2.2 One Quantitative Variable: Shape, Center, and Spread

- Goal: We need to turn “raw” data into information.
- Data are summarized into
 - 1 tables & graphs
 - 2 numerical indices
- One quantitative variable
 - 1 Visualization: dotplot, frequency table, and histogram
 - 2 Shape: symmetric or skewed
 - 3 Measures of center: mean and median
 - 4 Outliers and resistance
 - 5 Measures of spread: standard deviation, range, and IQR
 - 6 z-score

Example 2.3 | World gross for all 2011 Hollywood movies

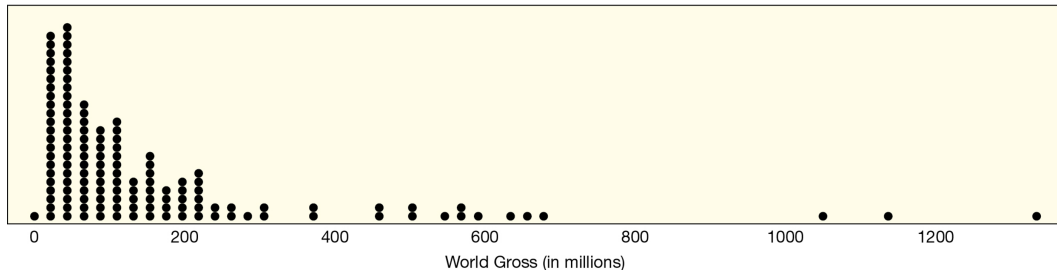
- We need to turn “raw” data into information.

	Movie	LeadStudio	RottenTomatoes	AudienceScore	Story	Genre	TheatersOpe
1	Insidious	Sony	67	65	Monster Force	Horror	2408
2	Paranormal Activity 3	Independent	68	58	Monster Force	Horror	3321
3	Bad Teacher	Independent	44	38	Comedy	Comedy	3049
4	Harry Potter and the Deathly Hallows Part 2	Warner Bros	96	92	Rivalry	Fantasy	4375
5	Bridesmaids	Relativity Media	90	77	Rivalry	Comedy	2918
6	Midnight in Paris	Sony	93	84	Love	Romance	944
7	The Help	Dreamworks Pictures	75	91	Maturation	Drama	2534
8	The Hangover Part II	Legendary Pictures	35	58	Comedy	Comedy	3615
9	Another Earth	Independent	63	74	Temptation	Fantasy	NA
10	Limitless	Virgin	69	73	Wretched Excess	Thriller	2756
11	Horrible Bosses	Warner Bros	69	72	Revenge	Comedy	3040
12	No Strings Attached	Spyglass Entertainment	49	57	Comedy	Comedy	3018
13	Twilight: Breaking Dawn	Independent	26	68	Love	Romance	4061
14	Transformers: Dark of the Moon	Dreamworks Pictures	35	67	Quest	Action	4088
15	Gnomeo and Juliet	Disney	56	52	Love	Animation	2994
16	Rio	20th Century Fox	71	73	Quest	Animation	3026
17	Super 8	Paramount	82	78	Monster Force	Horror	3379
18	Rise of the Planet of the Apes	20th Century Fox	83	87	Revenge	Action	3648
19	Apollo 18	Weinstein Company	23	31	Monster Force	Horror	3328
20	The Smurfs	Sony Pictures Animation	23	50	Fish Out Of water	Animation	3395
21	Fast Five	Universal	78	83	Escape	Action	3644
22	Our Idiot Brother	The Weinstein Company	68	79	Comedy	Comedy	2555
23	50/50	Independent	93	93	Discovery	Comedy	2458
24	Drive	Independent	93	79	Rivalry	Thriller	2886
25	Beginners	Independent	84	80	Love	Comedy	NA
26	Kung Fu Panda 2	Dreamworks Animation	82	80	Rivalry	Animation	3925
27	Unknown	Independent	55	57	The Riddle	Thriller	3043
28	The Ides of March	Columbia	85	76	Transformation	Thriller	2199

Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Dotplots

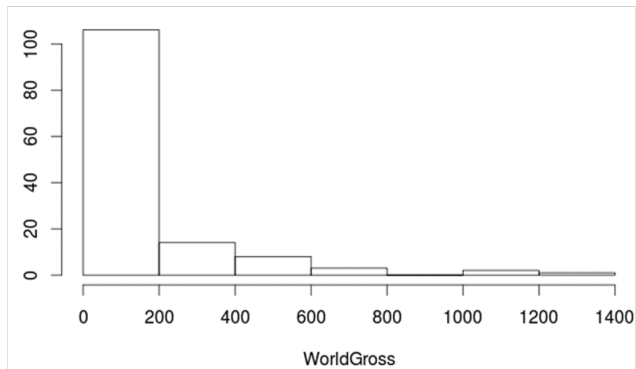
- In a **dotplot**, each case is represented by a dot and dots are stacked.
 - 1 Not only a graph that looks much like a histogram, but also a graph that shows **each case**. This is an advantage over histogram.
 - 2 The disadvantage is that for large data sets, the dotplot can be more unwieldy than histogram.
- Example 2.3: Hollywood movies 2011



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Histograms

- In a **frequency histogram**, the height of the each bar corresponds to the **number of cases** within that range of the variable.
- Example 2.3: Hollywood movies 2011



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Bar graph vs Histogram

- Bar graphs and pie charts are used to display frequency information of a categorical (qualitative) variable.
- Histograms are appropriate for displaying frequency information of a quantitative variable.
- Make sure every case falls into only one category for bar graphs and pie charts and only one subinterval for histograms.
- For a categorical variable, the number of bars equals the number of categories, and the number in each category is fixed.
- For a quantitative variable, the number of bars in a histogram is up to you (or your software), and the appearance can differ with different number of bars.

Example 2.4 | Antibiotic and weight gain

- An animal scientist is carrying out an experiment to investigate whether adding an antibiotic to the diet of chicks will promote growth over the diet without the antibiotic. The scientist feeds 100 chicks over eight weeks, and the weight gains for the 100 chicks are recorded as follows:

3.7	4.2	4.4	4.4	4.3	4.2	4.4	4.8	4.9	4.4
4.2	3.8	4.2	4.4	4.6	3.9	4.3	4.5	4.8	3.9
4.7	4.2	4.2	4.8	4.5	3.6	4.1	4.3	3.9	4.2
4.0	4.2	4.0	4.5	4.4	4.1	4.0	4.0	3.8	4.6
4.9	3.8	4.3	4.3	3.9	3.8	4.7	3.9	4.0	4.2
4.3	4.7	4.1	4.0	4.6	4.4	4.6	4.4	4.9	4.4
4.0	3.9	4.5	4.3	3.8	4.1	4.3	4.2	4.5	4.4
4.2	4.7	3.8	4.5	4.0	4.2	4.1	4.0	4.7	4.1
4.7	4.1	4.8	4.1	4.3	4.7	4.2	4.1	4.4	4.8
4.1	4.9	4.3	4.4	4.4	4.3	4.6	4.5	4.6	4.0

- Note:** Not organized at all! We need to turn “raw” data into information.

Example 2.4 | Summary using a frequency table

- Steps of constructing a frequency (or relative frequency) table (refer to Chapter 0 vides)
 - ① Find the **maximum** and **minimum** values: The largest weight gain is 4.9, and the smallest is 3.6.
 - ② Find the **range** of data: The range of data is $\text{max} - \text{min} = 4.9 - 3.6 = 1.3$.
 - ③ Select a **tentative number of subintervals**: No. of subintervals $K < \sqrt{n} = \sqrt{100} = 10$. Tentatively, let's use $K = 6$ subintervals.
 - ④ Select a **tentative width of each subinterval**: For each of 6 subintervals, an equal width of 0.2 may be a good choice ($\frac{\text{range}}{K} = \frac{1.3}{6} \approx 0.2$).
 - ⑤ Select a **tentative first subinterval** to include the minimum value: To include the smallest value 3.6, it may be convenient to choose the first interval to be $[3.55, 3.75)$.
 - ⑥ Expand the tentative first interval to a **last interval** to include the maximum value: Using the tentative width 0.2, expand the first subinterval until the resulting last interval to include the maximum value 4.9. After this, we have $[3.55, 3.75), [3.75, 3.95), \dots, [4.75, 4.95)$.
 - ⑦ **Refine the tentative subintervals**: However, the subintervals $[3.55, 3.75), [3.75, 3.95), \dots, [4.75, 4.95)$ are **not easy** to follow. A better choice for the subintervals could be $[3.50, 3.75), [3.75, 4.00), \dots, [4.75, 5.00)$.
 - ⑧ Find the **frequency (or relative frequency)** of cases falling in each subinterval.

Example 2.4 | Summary using a frequency table

- A “good” frequency (relative frequency) table of the antibiotic and weight gain data:

Subinterval	Frequency	Relative Frequency
$3.50 \leq - < 3.75$	2	$2/100 = 0.02$
$3.75 \leq - < 4.00$	22	$22/100 = 0.22$
$4.00 \leq - < 4.25$	24	$24/100 = 0.24$
$4.25 \leq - < 4.50$	30	$30/100 = 0.30$
$4.50 \leq - < 4.75$	13	$13/100 = 0.13$
$4.75 \leq - < 5.00$	9	$9/100 = 0.09$
Total	100	1.00

Making a Histogram

- Using the frequency table, we make a frequency (or relative frequency) histogram.
- Steps for making a histogram:
 - 1 Label **frequencies** on one axis and **subintervals** on the other axis.
 - 2 Draw a **rectangle** over each subinterval with a height equal to the **frequency (or relative frequency)** in the subinterval.
 - 3 Do **not** leave any space between two adjacent subintervals.

Example 2.4 | Summary using a histogram

- Make a frequency (or relative frequency) histogram of the antibiotic and weight gain data:
- Some examples of interpretation
 - 1 In the frequency table and the histogram, 76 of the 100 weight gain measurements are greater than 4 grams.
 - 2 The proportion (percentage) of the total measurements falling in those subintervals above 4 grams is 0.76 (76%).

Three Features of Data

■ What look like the data?

- 1 The **shape** of the distribution of the data.
- 2 While there are many different variety of shapes, certain shapes are common enough to have been named.

■ Where is the center of the data?

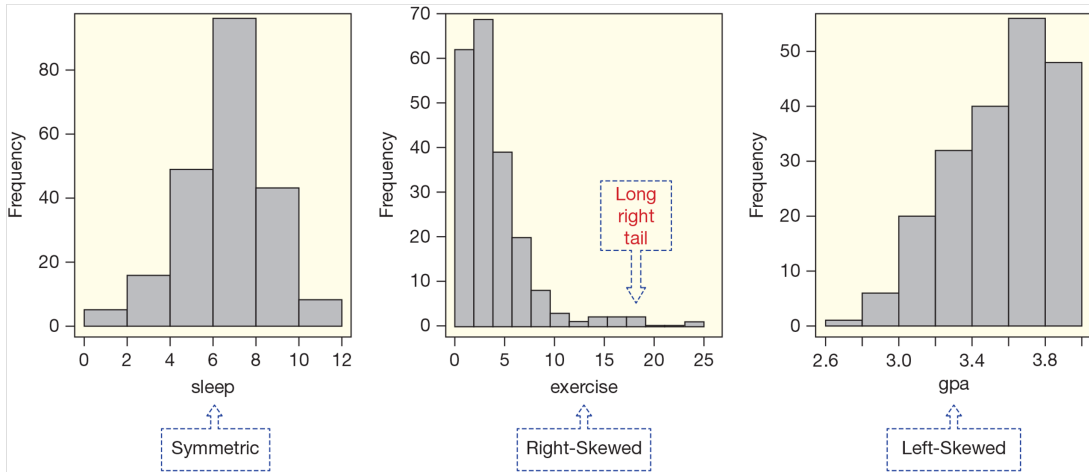
- 1 The **central location** of the distribution of the data is the most important aspect to notice and describe.
- 2 Measures of central location: **Mean**, **Median**, **Mode**.

■ How spread out the data?

- 1 The **variability** of the distribution of the data is the second most important feature.
- 2 Measures of spread: **Standard deviation**, **IQR**, **Range**.

Shapes of the Data (Symmetric or Skewed)

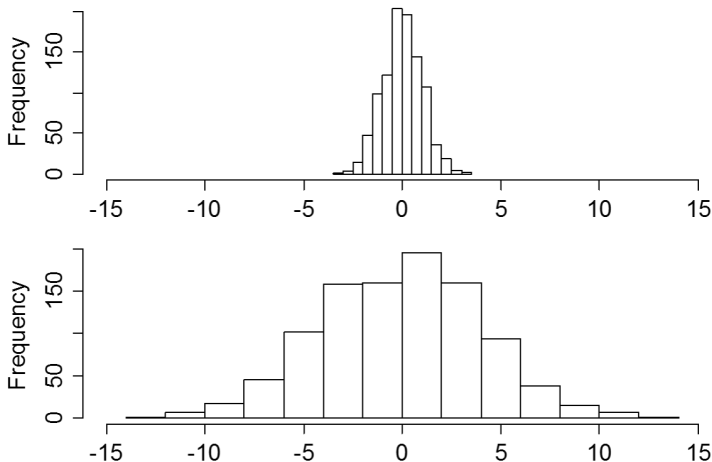
- Three common shapes are **symmetric**, **right-skewed**, and **left-skewed**.



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Bell-Shaped

- If the data distribution is symmetric, it could be **bell-shaped** (or **mound-shaped**).



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Meanings of Center

- The central location of the data may vary depending on:
 - 1 where is the **center of gravity** or **balance point**?
 - 2 where is the **middle** value?
 - 3 Where is the **most frequently occurring** value?

Mean | Definition

- The **mean** (or **average**) of a sample, denoted by $\bar{x}$, is

$$\bar{x} = \frac{\text{sum of all data values}}{\text{number of all data values}} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum x_i}{n}$$

- 1 x_i stands for the value from the i -th case.
 - 2 n denotes the number of cases in the sample (i.e., the sample size).
 - 3 $\sum$ (capital Greek sigma) indicates that a sum is to be taken.
 - 4 $\sum x_i$ denotes the sum of all the values $x_1, \dots, x_n$.
- Sample mean vs. population mean
 - 1 The **sample mean** $\bar{x}$ is the mean of all values in a **sample**.
 - 2 The **population mean** μ is the mean of all values in the **entire population**.

Example 2.5 | Sample mean and deviations

- A sample of $n = 10$ is taken as follows:

x_i	4	2	3	6	4	6	5	1	7	2
-------	---	---	---	---	---	---	---	---	---	---

- ① We compute the **sample mean $\bar{x}$** : using $x_1 = 4, x_2 = 2, \dots, x_9 = 7, x_{10} = 2$,

$$\bar{x} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_9 + x_{10}}{10} = \frac{4 + 2 + \dots + 7 + 2}{10} = \frac{40}{10} = 4$$

- ② Let's compute $x_i - \bar{x}$, the **deviation of x_i from the sample mean $\bar{x}$** , by subtracting $\bar{x}$ from each value x_i : $4 - 4 = 0, 2 - 4 = -2, \dots, 7 - 4 = 3, 2 - 4 = -2$.

$x_i - \bar{x}$	0	-2	-1	2	0	2	1	-3	3	-2
-----------------	---	----	----	---	---	---	---	----	---	----

- ③ Now, check if the sum of all deviations is 0:

$$\sum (x_i - \bar{x}) = 0 + (-2) + \dots + 3 + (-2) = 0,$$

which is **not a coincidence**.

Sample Mean | Properties

■ Properties of the sample mean $\bar{x}$

- 1 This is what most people think of as “the average.”
- 2 The **mean** is a useful measure for the central value of a set of measurements in the sense of mathematical reasoning and computation: it is a **center of gravity**, or **balancing point**, of the values.
- 3 The sample mean $\bar{x}$ locates itself so that $\sum(x_i - \bar{x}) = 0$.
- 4 The sample mean $\bar{x}$ can be very **SENSITIVE** to one or more **EXTREME** values in the data.

Median

- The **median** of a sample is the **middle value** when all the values are arranged **from smallest to largest**.
 - 1 The **sample median** is denoted by m or $\tilde{x}$.
 - 2 The **population median** is denoted by M or $\tilde{\mu}$.
- The median m divides the data into two groups: approximately **50%** of all the values lie **above** the median and **50%** fall **below** the median.
- How to find the sample median m ? After the values in a sample are **rearranged by the ascending order**, we find the median m as follows:
 - 1 for an **odd** sample size, m is just the **single middle value**.
 - 2 for an **even** sample size, m is the **mean of the two middle values**.

Example 2.6 | Sample median

- The following measurements are observed:

4.4	4.2	4.4	4.8	4.9	4.8	4.5	4.3	4.7
-----	-----	-----	-----	-----	-----	-----	-----	-----

- To find the sample median m , we first need to rearrange the data by the **ascending order**:

4.2	4.3	4.4	4.4	4.5	4.7	4.8	4.8	4.9
-----	-----	-----	-----	-----	-----	-----	-----	-----

- Second, because the sample size $n = 9$ is **odd**, there is **only one middle value**, so that we find the 5th value as the sample median: $m = 4.5$.

- If a new value 5.6 is included into the data, what is the new sample median?

- With the new value 5.6 included, the rearranged data is

4.2	4.3	4.4	4.4	4.5	4.7	4.8	4.8	4.9	5.6
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

- Because $n = 10$ is **even**, there are **two middle values** in the 5th and 6th. Then, the sample median is the **average of these two middle values**: $m = \frac{4.5+4.7}{2} = 4.6$.
- Caution**: Do **not** say there are two sample medians.

Exercise 2.2 | Measures of Center (Mean and Median)

- For each of the following samples, find the **mean** and the **median**, and identify any **outliers**.

1 8, 12, 3, 18, 15

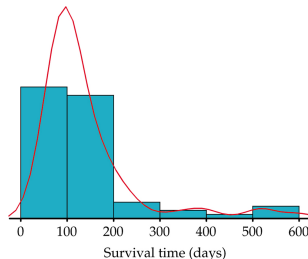
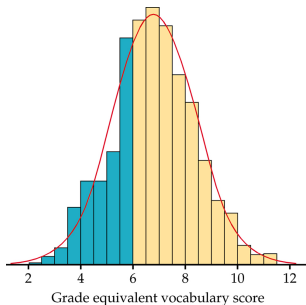
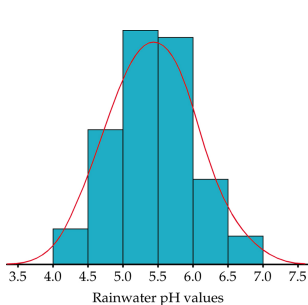
2 41, 53, 38, 12, 115, 47, 50

3 15, 22, 12, 28, 58, 18, 25, 18

4 110, 112, 118, 119, 122, 125, 129, 135, 138, 140

Mean and Median for Symmetrical Distributions

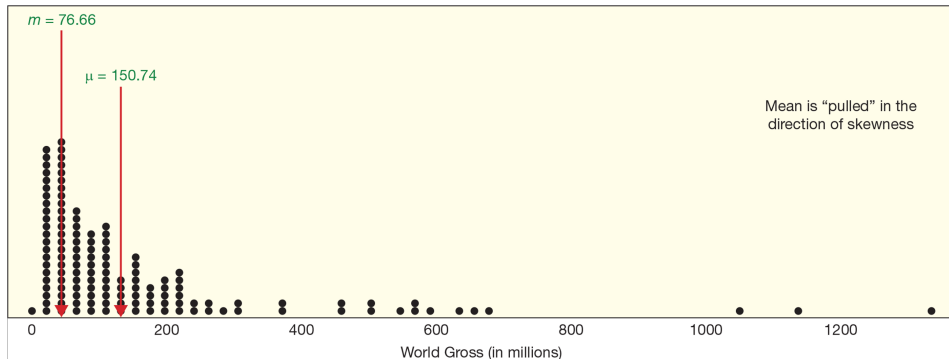
- Locations of the mean and median in bell-shaped and symmetric distributions:
 - 1 If a distribution is **symmetrical**, the lower half of the distribution is a mirror image of the upper half.
 - 2 If the distribution is **symmetric** (or **bell-shaped** or **mound-shaped**), the mean and the median will be the same.



Mean and Median for Skewed Distributions

- The mean is “pulled” in the direction of skewness.

① For a right-skewed distribution, the mean is expected to be higher than the median.

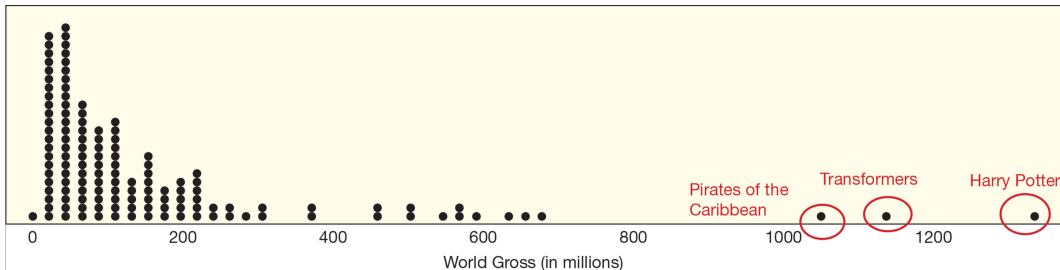


Source: Lock⁵ (2021), John Wiley & Sons, Inc.

② For a left-skewed distribution, the mean is expected to be lower than the median.

Outliers

- An **outlier** is an observed value that is **notably distinct** from the other values in a dataset.



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Resistance

- A statistic is called **resistant** (or **robust**) if it is relatively unaffected by outliers (or extreme values).
- The median is resistant while the mean is not.
- Example: World gross for the Hollywood movies 2011 dataset

	Mean	Median
With Harry Potter	\$150,742,300	\$76,658,500
Without Harry Potter	\$141,889,900	\$75,009,000

Example 2.7 | Sample mean and outliers

- A sample of $n = 15$ overdue accounts in a department store yields the following amounts due (in dollars):

55.20	4.88	271.95	18.06	180.29
365.29	28.16	399.11	807.80	44.14
97.47	9.98	61.61	56.89	82.73

- The sample mean is computed as

$$\bar{x} = \frac{\sum x_i}{n} = \frac{2,483.56}{15} = 165.57 \quad (\text{in dollars})$$

- However, **without the largest value 807.80**, the sample mean is now **substantially reduced** to

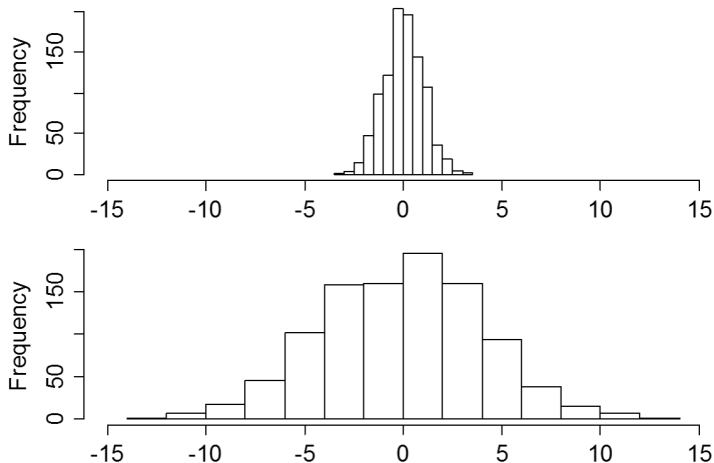
$$\bar{x} = \frac{\sum x_i}{n} = \frac{1,675.76}{14} = 119.70 \quad (\text{in dollars})$$

What to Do with Outliers?

- When using statistics that are **not resistant** to outliers, stop and think about whether the outlier is a mistake.
- If not, we have to decide whether the outlier is part of our population of interest or not.
- Usually, for outliers that are not a mistake, it's best to run the analysis twice, once with the outlier(s) and once without, to see how much the outlier(s) are affecting the results.
- In addition, we can apply statistical methods based on a mixture distribution.

Why Consider Spread?

- How are the two histograms **similar**, and how are they **different**?



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Example 2.8 | Spread

- Consider the following data sets:

Data set A	10	10	10	10	10	10	$\bar{x} = 10, m = 10$
Data set B	5	5	5	15	15	15	$\bar{x} = 10, m = 10$
Data set C	0	0	0	20	20	20	$\bar{x} = 10, m = 10$
Data set D	7	8	9	11	12	13	$\bar{x} = 10, m = 10$
Data set E	0	10	10	10	10	20	$\bar{x} = 10, m = 10$

- The five data sets have the same mean.
 - The five data sets have the same median.
- Because of these reasons, they can be considered similar?

Example 2.8 | Spread

- Consider the following data sets:

Data set A	10	10	10	10	10	10	$\bar{x} = 10, m = 10$
Data set B	5	5	5	15	15	15	$\bar{x} = 10, m = 10$
Data set C	0	0	0	20	20	20	$\bar{x} = 10, m = 10$
Data set D	7	8	9	11	12	13	$\bar{x} = 10, m = 10$
Data set E	0	10	10	10	10	20	$\bar{x} = 10, m = 10$

- Rank the previous five data sets in terms of variability (in your opinion) (*Answers may vary*):

- 1 = least variable = Data set ()
- 2 = = Data set ()
- 3 = = Data set ()
- 4 = = Data set ()
- 5 = most variable = Data set ()

Measures of Spread

- Three most popular measures of spread:
 - 1 Standard deviation
 - 2 Interquartile range (IQR)
 - 3 Range

Standard Deviation | Definition

- A **deviation of x_i from the mean $\bar{x}$** , denoted by $x_i - \bar{x}$, is the difference between the individual x_i and the mean $\bar{x}$.
- The **sample variance s^2** is

$$s^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1} = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

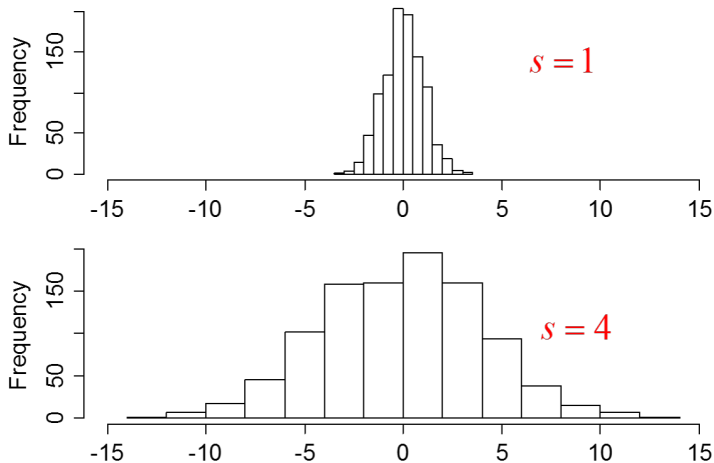
- The **sample standard deviation s** is $s = \sqrt{s^2}$.
- Sample standard deviation vs. population standard deviation
 - 1 The **sample standard deviation s** is the standard deviation of all values in a **sample**.
 - 2 The **population standard deviation σ** is the standard deviation of all values in the **entire population**.

Standard Deviation | Properties

- The standard deviation s gives a rough estimate of the typical distance of data values from the mean $\bar{x}$.
- $s \geq 0$, because $s^2 \geq 0$.
- The more variability there is in the data and the more spread out the data are from the center, the larger the standard deviation s (or s^2) is.
- If the data are constant (show no variability), then $s = 0$ (or $s^2 = 0$).
- The standard deviation s is measured in the same units as the data, while the variance s^2 is measured in squared units.
 - For example, if the data are in inches, then s is in inches but s^2 is in square inches.

Standard Deviation | Graphical comparison

- Approximate values of the sample standard deviation s :



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Example 2.8 | Standard deviation computation

- Compute the sample variance and sample standard deviation of the data set D.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
7	$7 - 10 = -3$	$(-3)^2 = 9$
8	$8 - 10 = -2$	$(-2)^2 = 4$
9	$9 - 10 = -1$	$(-1)^2 = 1$
11	$11 - 10 = 1$	$1^2 = 1$
12	$12 - 10 = 2$	$2^2 = 4$
13	$13 - 10 = 3$	$3^2 = 9$
Total	0	28

We obtain $\sum (x_i - \bar{x})^2 = 28$. The sample variance s^2 is

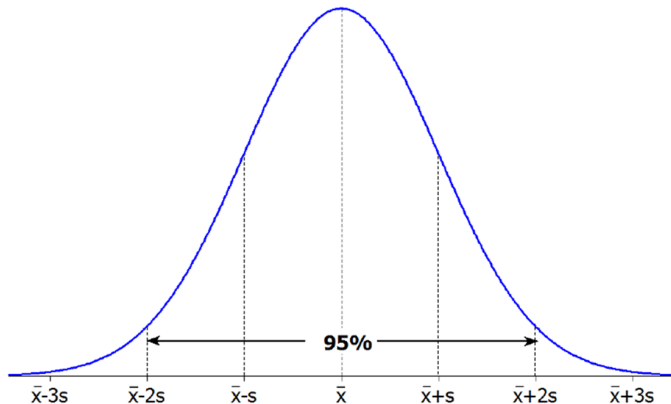
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \frac{28}{6 - 1} = \frac{28}{5} = 5.6$$

and the sample standard deviation is $s = \sqrt{s^2} = \sqrt{5.6} = 2.366$.

- Question: How do we interpret “ $s = 2.366$ ”?

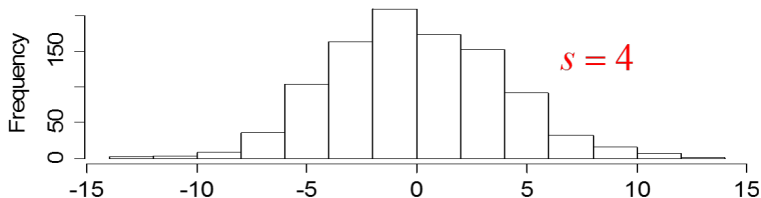
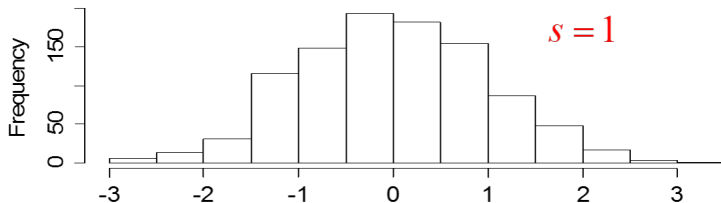
Standard Deviation | Empirical Rule

- **Empirical Rule:** This applies to “mound-shaped” or “bell-shaped” histograms only, which has a single peak, is symmetrical about the median, and tapers off gradually in the tails.
 - 1 The interval $[\bar{x} - s, \bar{x} + s]$ contains approximately 68% of all values.
 - 2 The interval $[\bar{x} - 2s, \bar{x} + 2s]$ contains approximately 95% of all values.
 - 3 The interval $[\bar{x} - 3s, \bar{x} + 3s]$ contains approximately almost all values.



Example 2.9 | Empirical Rule

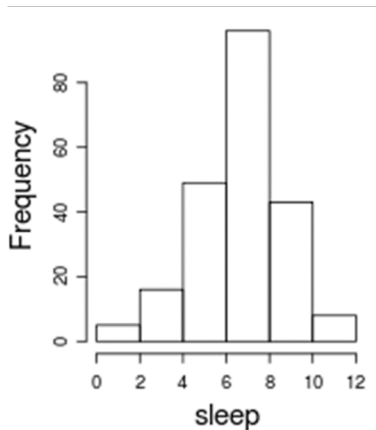
- Based on the histograms below, estimate the values of the sample standard deviations.



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Exercise 2.3 | Empirical Rule

- Based on the histogram below, estimate the value of the sample standard deviation.



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Example 2.10 | Empirical Rule

- A survey shows that pulse rates (in bpm) for a sample of college students are approximately bell-shaped with mean 69.6 and standard deviation 12.2. What interval is likely to contain 95% of the pulse rates in this sample?

Answer: We find

$$\begin{aligned} & 69.6 \pm 2(12.2) \\ &= 69.6 \pm 24.4 \\ &= 45.2 \text{ to } 94.0 \text{ bpm} \end{aligned}$$

Exercise 2.4 | Empirical Rule

- The average clean-up time for a crew of a medium-size firm is 84.0 hours and the standard deviation is 6.8 hours. Assume the empirical rule is appropriate.
 - ① Within what interval will the total clean-up time fall 95% of the time?
 - ② What proportion of the time will it take the clean-up crew 97.6 or more hours to clean the plant?

Standardization and Z-scores

- The **standardized value** (or **z-score**) for a data value x is obtained by

$$z = \frac{x - \bar{x}}{s}$$

- 1 For a population, $\bar{x}$ is replaced with μ and s is replaced with σ .
- 2 z-scores are **unitless**.
- 3 A z-score measures (i) **how many standard deviations** a data value x **falls away from the mean**, and (ii) **in which direction**.
- 4 When a data value x is larger than the mean, the z-score of x is positive. When x is smaller than the mean, the z-score of x is negative.
- 5 We often standardize observations to express and compare them in a **common** scale.
- 6 For bell-shaped distributions, 95% of all z-scores fall between -2 and 2 .

Example 2.11 | Z-score

- The heights of young women are approximately normal with mean $\mu = 65$ inches and $\sigma = 2$ inches. The standardized height is

$$z = \frac{\text{height} - \mu}{\sigma}$$

- 1 A women of 67 inches tall has the standardized height

$$z = \frac{\text{height} - \mu}{\sigma} = \frac{67 - 65}{2} = 1,$$

or 1 standard deviation above the mean height.

- 2 A women of 62 inches tall has the standardized height

$$z = \frac{\text{height} - \mu}{\sigma} = \frac{62 - 65}{2} = -1.5,$$

or 1.5 standard deviation below the mean height.

Exercise 2.5 | Z-score

- Three landmarks of baseball achievement are Ty Cobb's batting average of .420 in 1911, Ted Williams's .406 in 1941, and George Brett's .390 in 1980. These batting averages cannot be compared directly because the distribution of major league batting averages has changed over the decades. The distributions are quite normal. While the mean batting average has been held roughly constant, the standard deviation has dropped over time as follows:

Decade	Mean	Std. dev.
1910s	.266	.0371
1940s	.267	.0326
1970s	.261	.0317

Compute the standardized batting averages for Cobb, Williams, and Brett to compare how far each stood above his peers.